

Early Education and Holistic Human Development for the Future: Age-Appropriate Self-Regulation and Metacognition Learning Skills in Vietnam

1. Introduction

Early childhood education is increasingly recognized as a critical foundation for holistic human development, with particular emphasis on fostering self-regulation and metacognitive skills that prepare children for lifelong learning and well-being (■ Robson et al., 2020; Korucu et al., 2022; Vu, 2021; Dinh & Robinson, 2023). Research consistently demonstrates that self-regulation—including behavioral, cognitive, and emotional dimensions—is strongly associated with academic achievement, social competence, and positive mental health outcomes across the lifespan (■ Robson et al., 2020; Korucu et al., 2022; Ursache et al., 2012; Day et al., 2022). Metacognitive skills, which involve planning, monitoring, and reflecting on one’s own learning processes, begin to emerge in preschool years and are essential for developing independent learners (Narengaowa & Tungalag, 2024; ■ Eberhart et al., 2024; Whitebread et al., 2009; ■ Marulis & Nelson, 2020). In Vietnam, educational reforms have shifted from teacher-centered to child-centered approaches, aiming to integrate holistic development into early education curricula (Vu, 2021; Dinh & Robinson, 2023). However, challenges remain in operationalizing these concepts within policy and practice, particularly regarding cultural adaptation and teacher preparedness (Braund & Timmons, 2021; Duong & Nguyet, 2024; ■ Cabus et al., 2023). This review synthesizes global and Vietnamese research on age-appropriate self-regulation and metacognition in early education, highlighting effective strategies, developmental trajectories, contextual influences, and gaps relevant to Vietnam’s future educational landscape.

Does early childhood education that emphasizes self-regulation and metacognition improve holistic human development outcomes?

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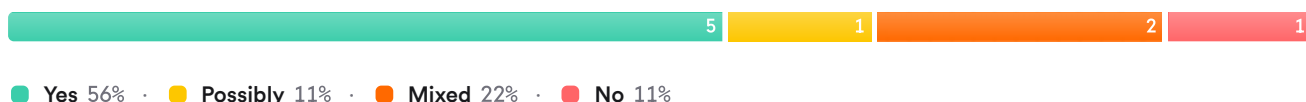


FIGURE 1 Consensus meter visualizing agreement on the impact of self-regulation/metacognition-focused early education on holistic development.

2. Methods

A comprehensive literature search was conducted across over 170 million research papers indexed in Consensus—including Semantic Scholar, PubMed, and other databases—using targeted queries related to early education, holistic development, self-regulation, metacognition, executive function, emotional competence, Vietnamese context, intervention strategies, critiques/barriers, and developmental outcomes. The search identified 1,091 potentially relevant papers; after de-duplication and screening for relevance to the topic (including both global frameworks and Vietnam-specific studies), 713 papers were screened. Of these, 596 met eligibility criteria based on relevance to age-appropriate self-regulation/metacognition in early childhood education. The final review included the top 50 most relevant papers.

Search Strategy

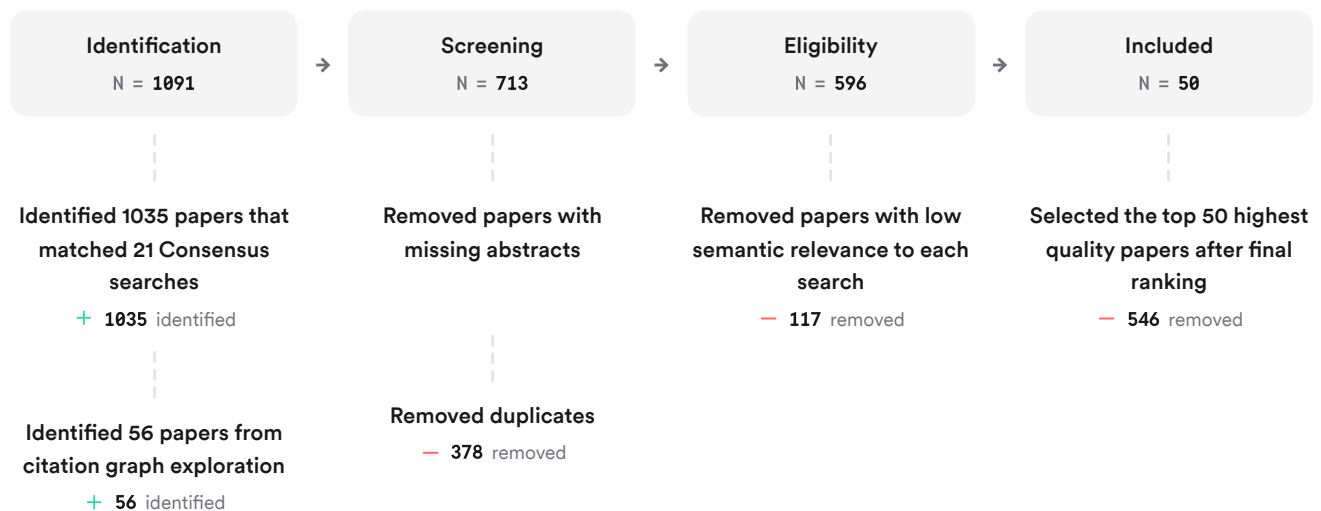


FIGURE 2 Flow diagram of literature search strategy from identification to inclusion.

Eight unique search groups were used to ensure comprehensive coverage of foundational theories, Vietnamese policy/culture context, intervention strategies, critiques/barriers, interdisciplinary perspectives (e.g., neuroscience), developmental breakdowns by age group, alternate terminology (e.g., executive function), and citation graph exploration.

3. Results

3.1. Theoretical Foundations of Self-Regulation & Metacognition

Self-regulation is a multifaceted construct encompassing behavioral control (inhibition/attention), cognitive regulation (working memory/executive function), and emotional regulation (Korucu et al., 2022; Ursache et al., 2012; Blair & Ku, 2022). These components develop rapidly between ages 3–7 but follow distinct trajectories: emotional regulation often precedes behavioral/cognitive regulation (Braund & Timmons, 2021; Korucu et al., 2022). Metacognitive skills—such as goal setting, monitoring progress, evaluating outcomes—emerge during preschool years but are initially limited; they become more sophisticated with age and adult scaffolding (Narengaowa & Tungalag, 2024; Goupil & Kouider, 2019; Whitebread et al., 2009). Integrated models emphasize the reciprocal relationship between executive function (cognitive control) and metacognition in supporting adaptive learning behaviors (Blair & Ku, 2022; Marulis et al., 2019).

3.2. Impact of Early Education on Holistic Development

Meta-analyses show that strong self-regulation in preschool predicts better academic performance (literacy/numeracy), social competence (peer relationships), mental health (lower internalizing/externalizing problems), healthier behaviors (reduced substance use/obesity), and even long-term life outcomes such as employment or criminality (■ Robson et al., 2020; Korucu et al., 2022; Ursache et al., 2012; Day et al., 2022). Holistic approaches that integrate cognitive, emotional, social domains yield optimal developmental gains (Rodiah et al., 2024; Gea et al., 2025), especially when curricula are tailored to children’s developmental stages.

3.3. Effective Strategies & Interventions

Teacher professional development programs targeting self-regulation/executive function result in significant gains for children compared to standard curricula (■ Muir et al., 2024; ■ Chen et al., 2024; Vasseleu et al., 2024). Interventions using play-based learning (■ Nakato et al., 2025), movement/physical activity (Savina, 2024; Wu et al., 2024), nature-based practices (Ernst & Stelley, 2024), mindfulness-based interventions (Bockmann & Yu, 2022), music/play integration (■ Zachariou et al., 2023), or dynamic assessment methodologies (Moreira et al., 2024) all demonstrate positive effects on self-regulation/metacognition. Teacher scaffolding is crucial: young children benefit most when adults model reflective thinking or provide structured opportunities for planning/monitoring/evaluating their own learning processes (Narengaowa & Tungalag, 2024; ■ Eberhart et al., 2024; Erdmann & Hertel, 2019).

3.4. Vietnamese Context: Policy Evolution & Implementation Challenges

Vietnam’s early childhood education system has undergone major reforms since the late 20th century—shifting toward child-centered pedagogy with explicit aims for holistic development (physical/emotional/intellectual/aesthetic) as outlined in national policy documents since 1998/2005 (Vu, 2021). Recent longitudinal studies confirm that preschool attendance in Vietnam is associated with persistent gains in vocabulary/mathematics/life satisfaction up to adolescence—even among disadvantaged groups—though effect sizes are modest (Dinh & Robinson, 2023). However, implementation challenges persist: teachers report limited understanding of self-regulation concepts or intervention strategies; professional training remains uneven; cultural values such as filial piety may shape socio-emotional goals differently than Western models; rural/ethnic minority children face additional barriers to access high-quality programs (Duong & Nguyet, 2024; ■ Cabus et al., 2023; Dinh & Robinson, 2023).

Results Timeline

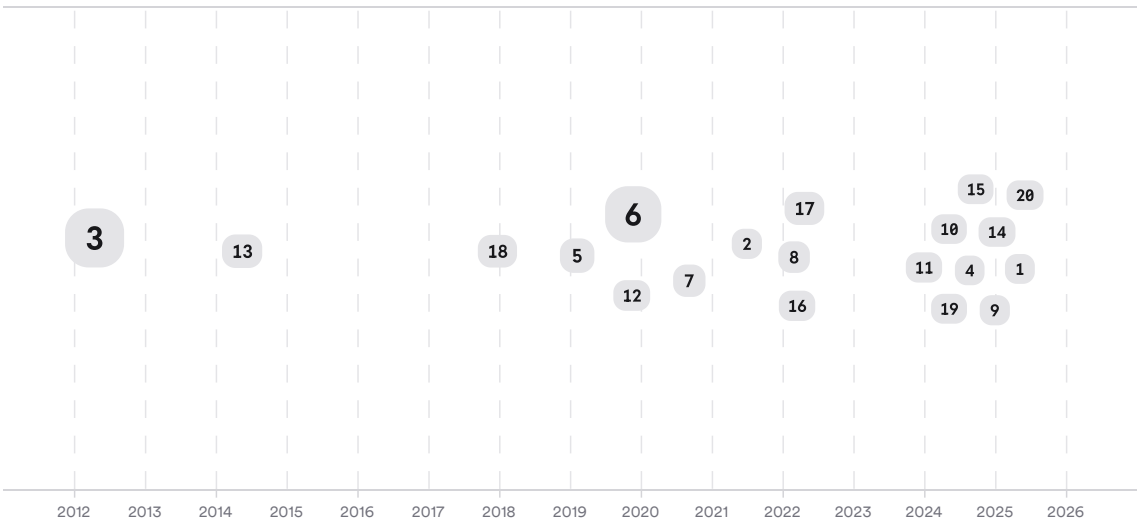


FIGURE 3 Timeline of key research findings on self-regulation/metacognition interventions in early childhood globally/Vietnam. Larger markers indicate more citations.

Top Contributors

Type	Name	Papers
Author	Steven J Howard	(■ Muir et al., 2024; Day et al., 2022; Vasseleu et al., 2024)
Author	E. Vasseleu	(Vasseleu et al., 2024; Vasseleu et al., 2021)
Author	D. Whitebread	(Erdmann & Hertel, 2019; Marulis et al., 2019; Whitebread et al., 2009)

Type	Name	Papers
Journal	<i>Frontiers in Psychology</i>	(Korucu et al., 2022; Blair & Ku, 2022; Fridman et al., 2020)
Journal	<i>Early Childhood Education Journal</i>	(Savina, 2020; Ross & Ascetta, 2024; Bockmann & Yu, 2022)
Journal	<i>Metacognition and Learning</i>	( Eberhart et al., 2024; Erdmann & Hertel, 2019)

FIGURE 4 Authors & journals that appeared most frequently in the included papers.

4. Discussion

The evidence base strongly supports prioritizing age-appropriate self-regulation and metacognitive skill development within early childhood education as a means of promoting holistic human development—academically as well as socially/emotionally—for both global populations and within Vietnam specifically ( Robson et al., 2020; Korucu et al., 2022; Vu, 2021; Dinh & Robinson, 2023). High-quality interventions are most effective when they combine explicit instruction with play-based or experiential learning environments tailored to children’s developmental stages; teacher training/professional support is essential for successful implementation at scale ( Muir et al., 2024;  Chen et al., 2024; Vasseleu et al., 2024).

However, conceptual ambiguity persists around definitions/measures of self-regulation/metacognition—especially across cultures—which complicates curriculum design/evaluation efforts (Braund & Timmons, 2021; Ross & Ascetta, 2024). In Vietnam’s context specifically: while national policies now emphasize holistic goals aligned with international best practices (Vu, 2021), practical barriers include uneven teacher preparation (Duong & Nguyet, 2024), resource constraints ( Cabus et al., 2023), rural/ethnic disparities (Dinh & Robinson, 2023), and potential misalignment between imported frameworks versus local cultural values.

Despite these challenges—and some critiques regarding over-individualization or neoliberal framing of “self-regulated” children ( Diaz-Diaz, 2022)—the overall weight of evidence indicates that fostering these skills is both feasible and beneficial if adapted thoughtfully to local contexts.

Claims & Evidence Table

Claim	Evidence Strength	Reasoning	Papers
Early self-regulation predicts academic/social/mental health outcomes	 Strong	Supported by large meta-analyses showing robust associations across multiple domains/life stages	( Robson et al., 2020; Korucu et al., 2022)
Metacognitive interventions improve young children's learning	 Strong	Multiple meta-analyses/intervention studies show moderate-to-large effects on SR/academic achievement	( Eberhart et al., 2024;  Chen et al., 2024)
Teacher professional development enhances classroom SR outcomes	 Strong	RCTs/mixed-methods studies show significant gains when teachers receive targeted SR/metacog training	( Muir et al., 2024; Vasseleu et al., 2024)
Holistic ECE models outperform narrow academic-only approaches	 Moderate	Systematic reviews/regression analyses show integrated cognitive/emotional/social focus yields best results	(Rodiah et al., 2024; Gea et al., 2025)

Claim	Evidence Strength	Reasoning	Papers
Implementation barriers limit impact in low-resource/rural settings	Moderate	Qualitative/Vietnamese studies highlight gaps due to resources/training/cultural adaptation	(Duong & Nguyet, 2024; Cabus et al., 2023)
Definitions/measures of SR/metacog lack cross-cultural clarity	Moderate	Reviews note conceptual/methodological inconsistencies hinder evaluation/comparison	(Braund & Timmons, 2021; Ross & Ascetta, 2024)

FIGURE 5 Key claims and support evidence identified in these papers.

5. Conclusion

In summary: Age-appropriate self-regulation and metacognitive skill-building are foundational pillars for holistic human development through early education—yielding benefits across academic achievement as well as social-emotional well-being into adulthood. While Vietnam’s evolving policies align with global best practices emphasizing these competencies (Vu, 2021), practical implementation requires ongoing investment in teacher training/professional support (Muir et al., 2024), culturally responsive curriculum design (Duong & Nguyet, 2024), equitable access for disadvantaged groups (Dinh & Robinson, 2023), and clearer conceptual frameworks (Braund & Timmons, 2021).

Research Gaps

Topic/Outcome	Preschoolers (<6y)	Primary students (6–12y)	Teachers/Educators	Rural/Ethnic Minorities
Self-Regulation	18	7	9	5
Metacognition	13	6	5	2
Intervention Effectiveness	10	4	7	1
Policy/Cultural Adaptation	3	1	4	3

FIGURE 6 Matrix showing concentration/gaps by topic/outcome vs population/study attribute.

Open Research Questions

Future research should focus on refining culturally appropriate definitions/measures of SR/metacognition; scaling up teacher professional development; evaluating long-term impacts across diverse Vietnamese populations; exploring how local values shape socio-emotional goals.

Question	Why
How can culturally adapted interventions best foster self-regulation/metacognition among rural or ethnic minority children	Addressing equity requires understanding how local culture/resources influence intervention effectiveness

Question	Why
in Vietnam?	
What are the long-term impacts of integrated SR/metacognitive curricula on life outcomes beyond primary school?	Longitudinal data will clarify persistence/fading of benefits into adolescence/adulthood
How can teacher professional development be optimized for sustainable classroom practice change?	Effective scale-up depends on sustainable models tailored to real-world constraints

FIGURE 7 Open questions guiding future research directions.

In conclusion: Prioritizing age-appropriate self-regulation/metacognitive skill-building within a holistic framework is essential for preparing Vietnamese children—and all learners—for a thriving future.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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